FairCKD: Fairness Audit Report

Investigating Bias and Fairness in Machine Learning Models for Automated CKD Diagnosis
M.Tech Dissertation Phase-II · Jimil Jingar · PIET, Parul University

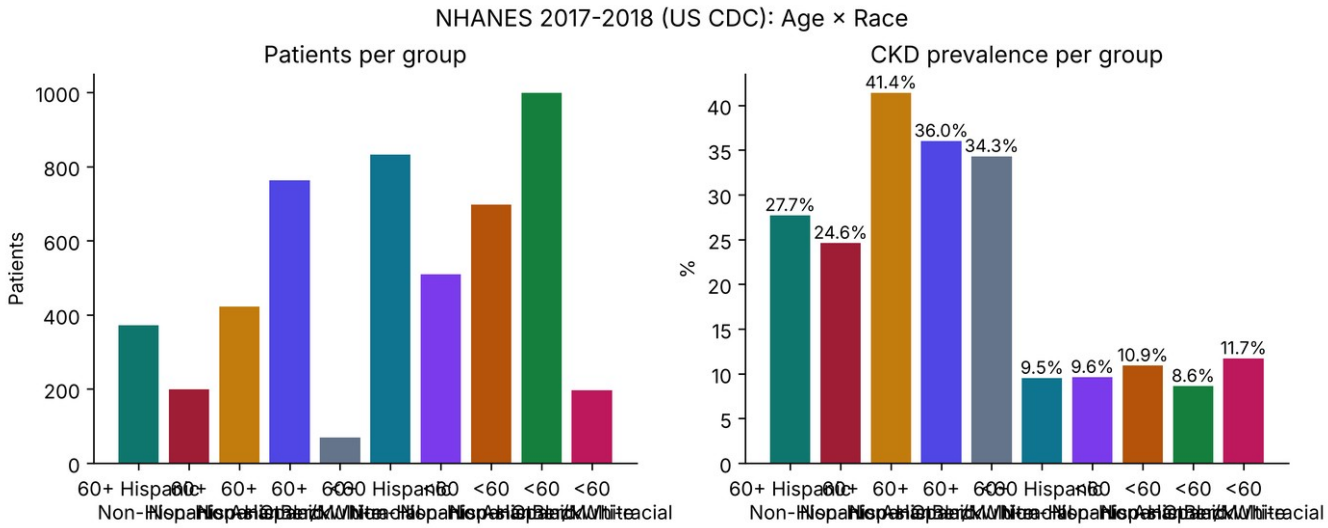
18 (US CDC)	
s (17 features, no kidney-function tests)	
Non-Hispanic Asian, 60+ Non-Hispanic Black, 60+ Non-Hispanic White, 60+ Other/Multi-racial, <60 Hispanic, <60 Non-Hispanic Asian, <60 Non-Hispanic Black, <60 No	
revalence	
epeats (baseline), 5 repeats (mitigation)	

Key findings

• Best baseline model: **Random Forest** (accuracy 0.738, balanced accuracy 0.736, AUC 0.807).
• Bias detection: in **3 of 7** models, CKD is detected least often in the **<60 Non-Hispanic White** group (Equal Opportunity gap 0.460–0.685).
• Even a perfect classifier has DP diff = 0.328 and DI = 0.208 here, because CKD prevalence differs between groups; error-rate metrics (Equal Opportunity, Equalized Odds) are the clinically meaningful fairness criteria.
• Logistic Regression: best mitigation (balanced accuracy loss ≤ 3 points) = **Unawareness**, Equalized Odds gap 0.622 → 0.541, balanced accuracy 0.736 → 0.730.
• XGBoost: best mitigation (balanced accuracy loss ≤ 3 points) = **Unawareness**, Equalized Odds gap 0.627 → 0.473, balanced accuracy 0.734 → 0.722.
• **FairCKD-Select** (proposed, nested CV): Equalized Odds gap 0.615 → **0.484** compared with accuracy-driven model selection, balanced accuracy 0.734 → 0.723.
• Statistical significance: 9 of 10 mitigation results reduce the Equalized Odds gap significantly (paired bootstrap, $p < 0.05$).
• External validation (NHANES 2015–16): baseline balanced accuracy 0.740–0.749, Equalized Odds gap 0.638–0.679; best mitigated gap 0.317.

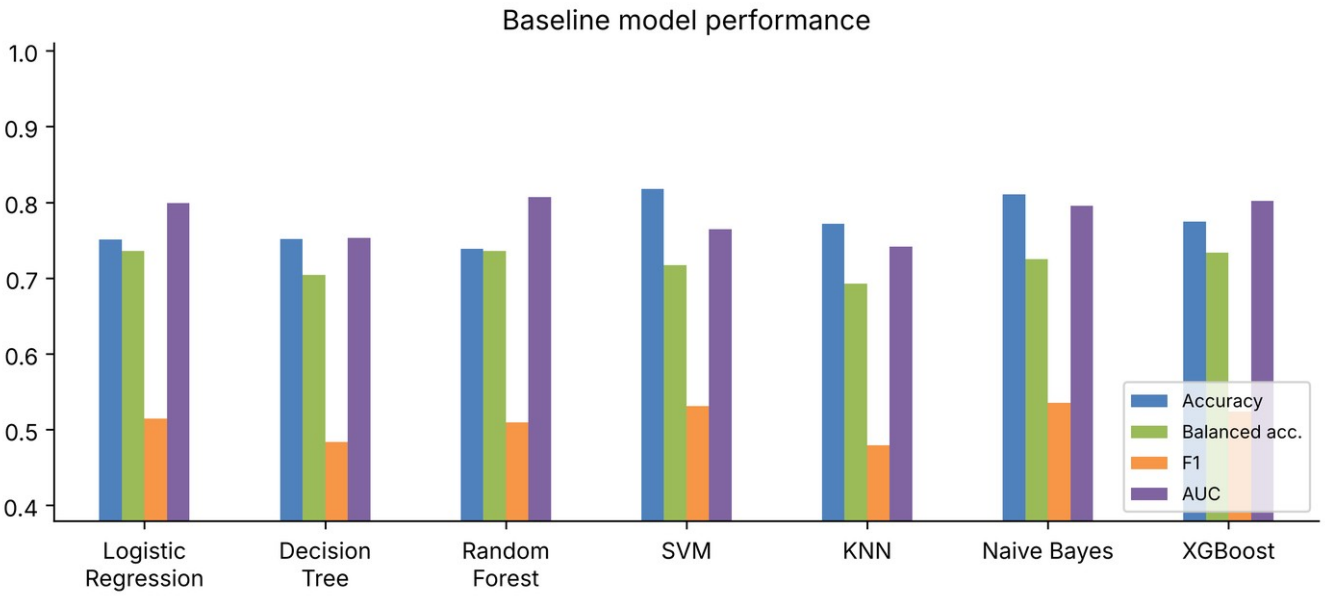
1. Dataset and groups

Group	Patients	Share (%)	CKD cases	CKD prevalence (%)	Missing values / patient
60+ Hispanic	372	7.3	103	27.7	0.160
60+ Non-Hispanic Asian	199	3.9	49	24.6	0.140
60+ Non-Hispanic Black	423	8.4	175	41.4	0.120
60+ Non-Hispanic White	763	15.1	275	36	0.130
60+ Other/Multi-racial	70	1.4	24	34.3	0.140
<60 Hispanic	832	16.4	79	9.5	0.210
<60 Non-Hispanic Asian	510	10.1	49	9.6	0.180
<60 Non-Hispanic Black	698	13.8	76	10.9	0.150
<60 Non-Hispanic White	999	19.7	86	8.6	0.130
<60 Other/Multi-racial	197	3.9	23	11.7	0.090
All	5063	100	939	18.5	0.150


NHANES 2017-2018 (US CDC): Age × Race
Patients per group
Patients
0
200
400
600
800
1000
CKD prevalence per group
%
0
5
10
15
20
25
30
35
40
27.7%
24.6%
41.4%
36.0%
34.3%
9.5%
9.6%
10.9%
8.6%
11.7%
60+ Hispanic
<60 Hispanic
<60
Non-Hispanic White

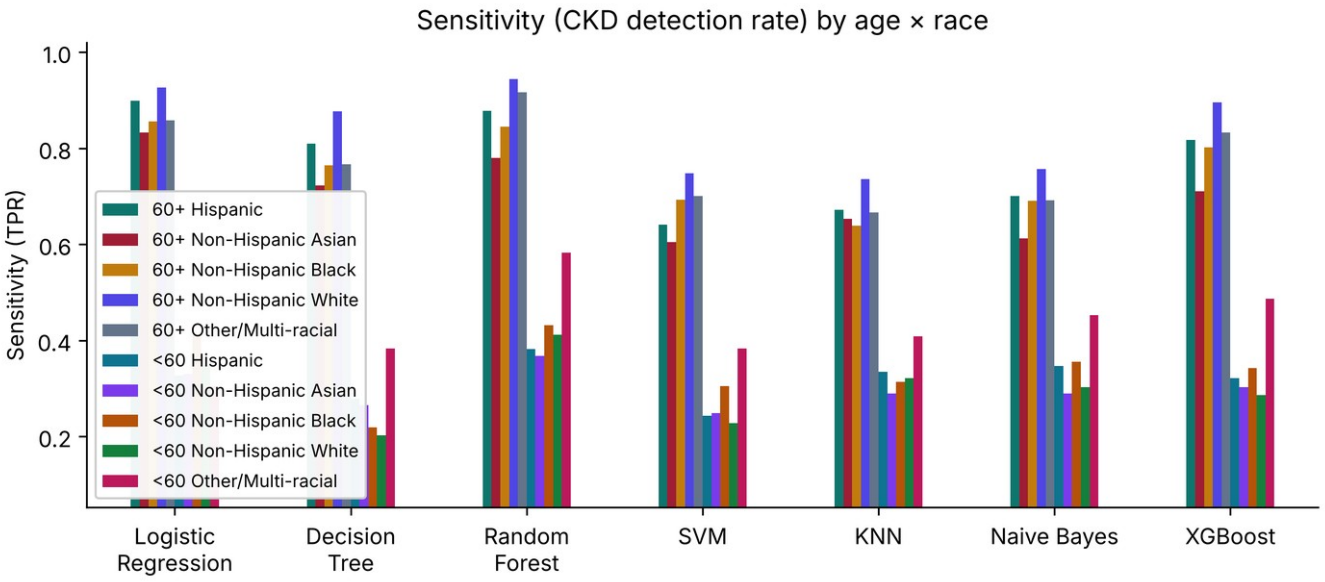
2. Baseline model performance

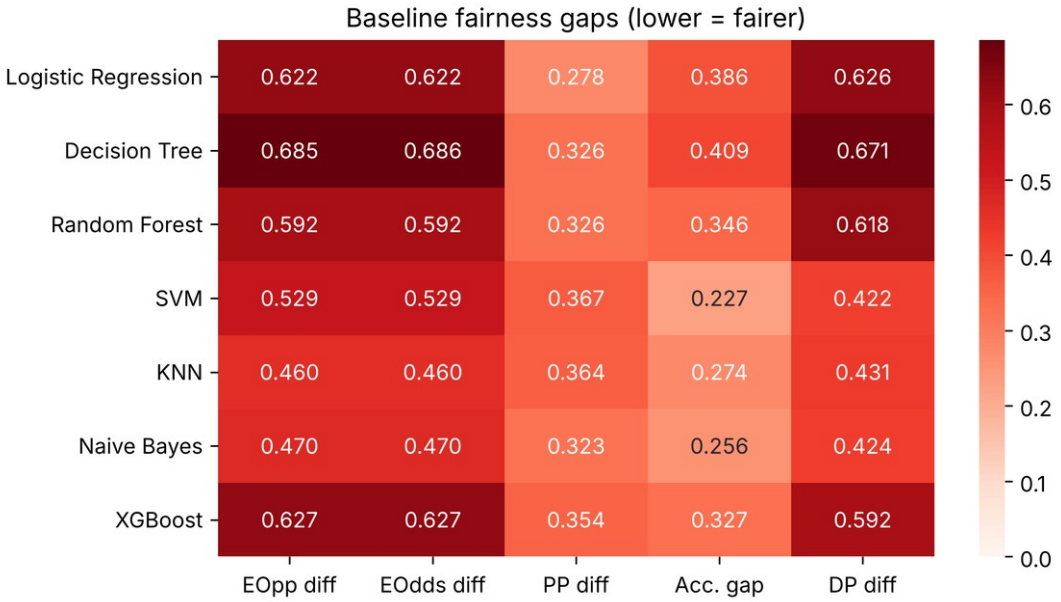
	Accuracy	Precision	Recall	F1	AUC	Balanced acc.
Logistic Regression	0.751 ± 0.002	0.403 ± 0.002	0.712 ± 0.004	0.514 ± 0.002	0.799 ± 0.000	0.736 ± 0.001
Decision Tree	0.751 ± 0.012	0.394 ± 0.013	0.628 ± 0.022	0.484 ± 0.009	0.753 ± 0.007	0.704 ± 0.007
Random Forest	0.738 ± 0.003	0.391 ± 0.004	0.732 ± 0.006	0.509 ± 0.005	0.807 ± 0.002	0.736 ± 0.004
SVM	0.818 ± 0.001	0.508 ± 0.003	0.556 ± 0.004	0.531 ± 0.003	0.764 ± 0.003	0.717 ± 0.002
KNN	0.772 ± 0.003	0.415 ± 0.005	0.567 ± 0.003	0.479 ± 0.003	0.742 ± 0.004	0.693 ± 0.002
Naive Bayes	0.811 ± 0.001	0.492 ± 0.003	0.588 ± 0.003	0.536 ± 0.002	0.795 ± 0.001	0.725 ± 0.002
XGBoost	0.774 ± 0.002	0.430 ± 0.003	0.669 ± 0.005	0.524 ± 0.003	0.802 ± 0.003	0.734 ± 0.002



3. Bias detection and fairness evaluation

	DP diff	DI ratio	EOpp diff	EOdds diff	PP diff	Acc. gap
Logistic Regression	0.626 ± 0.007	0.137 ± 0.007	0.622 ± 0.006	0.622 ± 0.006	0.278 ± 0.011	0.386 ± 0.009
Decision Tree	0.671 ± 0.045	0.100 ± 0.010	0.685 ± 0.043	0.686 ± 0.042	0.326 ± 0.020	0.409 ± 0.031
Random Forest	0.618 ± 0.004	0.176 ± 0.005	0.592 ± 0.024	0.592 ± 0.024	0.326 ± 0.012	0.346 ± 0.015
SVM	0.422 ± 0.006	0.122 ± 0.008	0.529 ± 0.020	0.529 ± 0.020	0.367 ± 0.020	0.227 ± 0.010
KNN	0.431 ± 0.012	0.194 ± 0.014	0.460 ± 0.021	0.460 ± 0.021	0.364 ± 0.031	0.274 ± 0.014
Naive Bayes	0.424 ± 0.015	0.148 ± 0.005	0.470 ± 0.010	0.470 ± 0.010	0.323 ± 0.016	0.256 ± 0.017
XGBoost	0.592 ± 0.011	0.121 ± 0.008	0.627 ± 0.018	0.627 ± 0.018	0.354 ± 0.018	0.327 ± 0.024
Perfect classifier (reference)	0.328	0.208	0.000	0.000	0.000	0.000


Sensitivity (CKD detection rate) by age × race
Sensitivity (TPR)
1.0
0.8
0.6
0.4
0.2
60+ Hispanic
60+ Non-Hispanic Asian
60+ Non-Hispanic Black
60+ Non-Hispanic White
60+ Other/Multi-racial
<60 Hispanic
<60 Non-Hispanic Asian
<60 Non-Hispanic Black
<60 Non-Hispanic White
<60 Other/Multi-racial
Logistic Regression
Decision Tree
Random Forest
SVM
KNN
Naive Bayes
XGBoost


Baseline fairness gaps (lower = fairer)
Logistic Regression 0.622 0.622 0.278 0.386 0.626
Decision Tree 0.685 0.686 0.326 0.409 0.671
Random Forest 0.592 0.592 0.326 0.346 0.618
SVM 0.529 0.529 0.367 0.227 0.422
KNN 0.460 0.460 0.364 0.274 0.431
Naive Bayes 0.470 0.470 0.323 0.256 0.424
XGBoost 0.627 0.627 0.354 0.327 0.592
EOpp diff
EOdds diff
PP diff
Acc. gap
DP diff
0.6
0.5
0.4
0.3
0.2
0.1
0.0

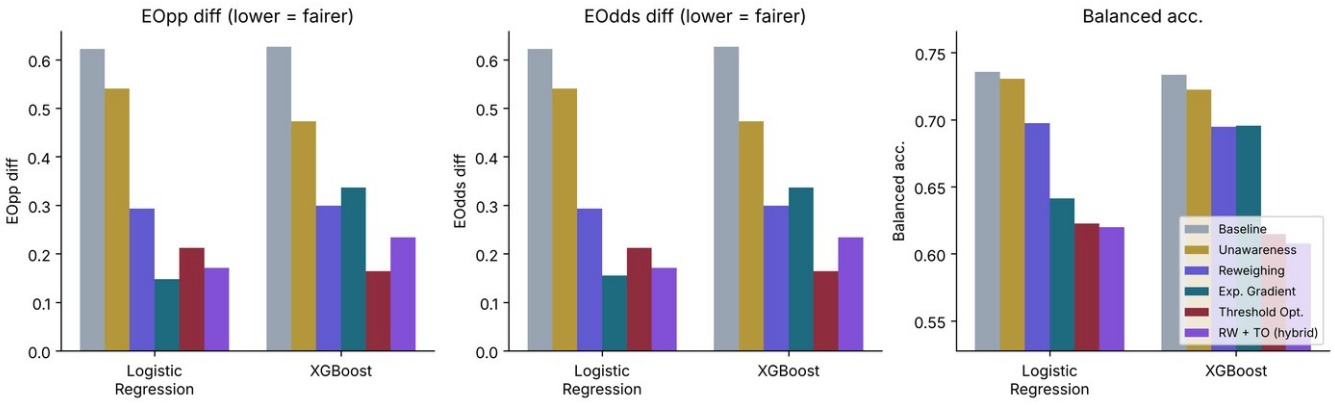
4. Bias mitigation

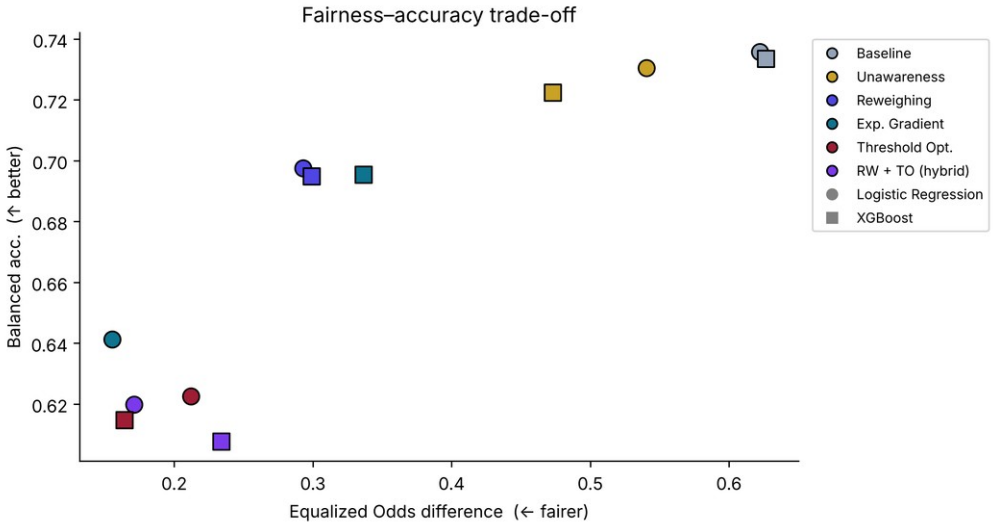
a) Performance (mean ± SD over CV repeats)

Model	Method	Accuracy	Balanced acc.	Recall	F1
Logistic Regression	Baseline	0.751 ± 0.002	0.736 ± 0.001	0.712 ± 0.004	0.514 ± 0.002
Logistic Regression	Unawareness	0.761 ± 0.001	0.730 ± 0.001	0.682 ± 0.003	0.514 ± 0.001
Logistic Regression	Reweighing	0.722 ± 0.001	0.698 ± 0.001	0.658 ± 0.003	0.468 ± 0.001
Logistic Regression	Exp. Gradient	0.677 ± 0.008	0.641 ± 0.008	0.585 ± 0.009	0.402 ± 0.009
Logistic Regression	Threshold Opt.	0.733 ± 0.013	0.623 ± 0.007	0.447 ± 0.029	0.383 ± 0.010
Logistic Regression	RW + TO (hybrid)	0.672 ± 0.026	0.620 ± 0.007	0.536 ± 0.050	0.378 ± 0.008
XGBoost	Baseline	0.774 ± 0.002	0.734 ± 0.002	0.669 ± 0.005	0.524 ± 0.003
XGBoost	Unawareness	0.771 ± 0.002	0.722 ± 0.002	0.645 ± 0.005	0.511 ± 0.003
XGBoost	Reweighing	0.748 ± 0.007	0.695 ± 0.007	0.611 ± 0.010	0.473 ± 0.010
XGBoost	Exp. Gradient	0.781 ± 0.002	0.695 ± 0.007	0.559 ± 0.014	0.487 ± 0.009
XGBoost	Threshold Opt.	0.702 ± 0.025	0.615 ± 0.011	0.476 ± 0.053	0.372 ± 0.013
XGBoost	RW + TO (hybrid)	0.694 ± 0.045	0.608 ± 0.004	0.470 ± 0.065	0.363 ± 0.004

b) Fairness gaps (lower = fairer; DI ratio: 1 = fair)

Model	Method	EOpp diff	EOdds diff	PP diff	DP diff	DI ratio	Acc. gap
Logistic Regression	Baseline	0.622 ± 0.006	0.622 ± 0.006	0.278 ± 0.011	0.626 ± 0.007	0.137 ± 0.007	0.386 ± 0.009
Logistic Regression	Unawareness	0.541 ± 0.018	0.541 ± 0.018	0.330 ± 0.010	0.503 ± 0.007	0.196 ± 0.002	0.310 ± 0.006
Logistic Regression	Reweighing	0.293 ± 0.012	0.293 ± 0.012	0.392 ± 0.009	0.314 ± 0.003	0.402 ± 0.004	0.188 ± 0.012
Logistic Regression	Exp. Gradient	0.147 ± 0.027	0.155 ± 0.021	0.464 ± 0.016	0.097 ± 0.027	0.767 ± 0.058	0.080 ± 0.015
Logistic Regression	Threshold Opt.	0.212 ± 0.058	0.212 ± 0.058	0.449 ± 0.018	0.104 ± 0.020	0.672 ± 0.034	0.142 ± 0.027
Logistic Regression	RW + TO (hybrid)	0.171 ± 0.038	0.171 ± 0.038	0.417 ± 0.003	0.116 ± 0.021	0.718 ± 0.049	0.084 ± 0.039
XGBoost	Baseline	0.627 ± 0.018	0.627 ± 0.018	0.354 ± 0.018	0.592 ± 0.011	0.121 ± 0.008	0.327 ± 0.024
XGBoost	Unawareness	0.473 ± 0.020	0.473 ± 0.020	0.400 ± 0.011	0.444 ± 0.023	0.215 ± 0.013	0.275 ± 0.017
XGBoost	Reweighing	0.299 ± 0.026	0.299 ± 0.026	0.508 ± 0.018	0.260 ± 0.012	0.430 ± 0.020	0.143 ± 0.021
XGBoost	Exp. Gradient	0.337 ± 0.040	0.337 ± 0.040	0.511 ± 0.015	0.261 ± 0.008	0.350 ± 0.013	0.187 ± 0.026
XGBoost	Threshold Opt.	0.164 ± 0.056	0.164 ± 0.056	0.428 ± 0.020	0.098 ± 0.017	0.718 ± 0.041	0.129 ± 0.020
XGBoost	RW + TO (hybrid)	0.234 ± 0.084	0.234 ± 0.084	0.430 ± 0.041	0.098 ± 0.017	0.715 ± 0.068	0.136 ± 0.044





c) Statistical significance (paired bootstrap vs. baseline, 1000 resamples; Δ = method – baseline)

Model	Method	Δ EOdds gap	95% CI (EOdds)	p (EOdds)	Δ Balanced acc.	95% CI (Bal. acc.)	p (Bal. acc.)
Logistic Regression	Unawareness	-0.099	[-0.213, -0.019]	0.030	-0.008	[-0.016, 0.000]	0.066
Logistic Regression	Reweighing	-0.314	[-0.467, -0.185]	0.000	-0.040	[-0.055, -0.026]	0.000
Logistic Regression	Exp. Gradient	-0.442	[-0.604, -0.286]	0.000	-0.087	[-0.106, -0.069]	0.000
Logistic Regression	Threshold Opt.	-0.440	[-0.625, -0.229]	0.002	-0.104	[-0.123, -0.086]	0.000
Logistic Regression	RW + TO (hybrid)	-0.441	[-0.618, -0.261]	0.000	-0.111	[-0.129, -0.093]	0.000
XGBoost	Unawareness	-0.113	[-0.220, 0.004]	0.060	-0.010	[-0.022, 0.002]	0.092
XGBoost	Reweighing	-0.318	[-0.447, -0.174]	0.000	-0.042	[-0.055, -0.028]	0.000
XGBoost	Exp. Gradient	-0.312	[-0.443, -0.193]	0.000	-0.045	[-0.060, -0.030]	0.000
XGBoost	Threshold Opt.	-0.402	[-0.578, -0.219]	0.000	-0.118	[-0.135, -0.100]	0.000
XGBoost	RW + TO (hybrid)	-0.400	[-0.583, -0.204]	0.000	-0.121	[-0.140, -0.103]	0.000

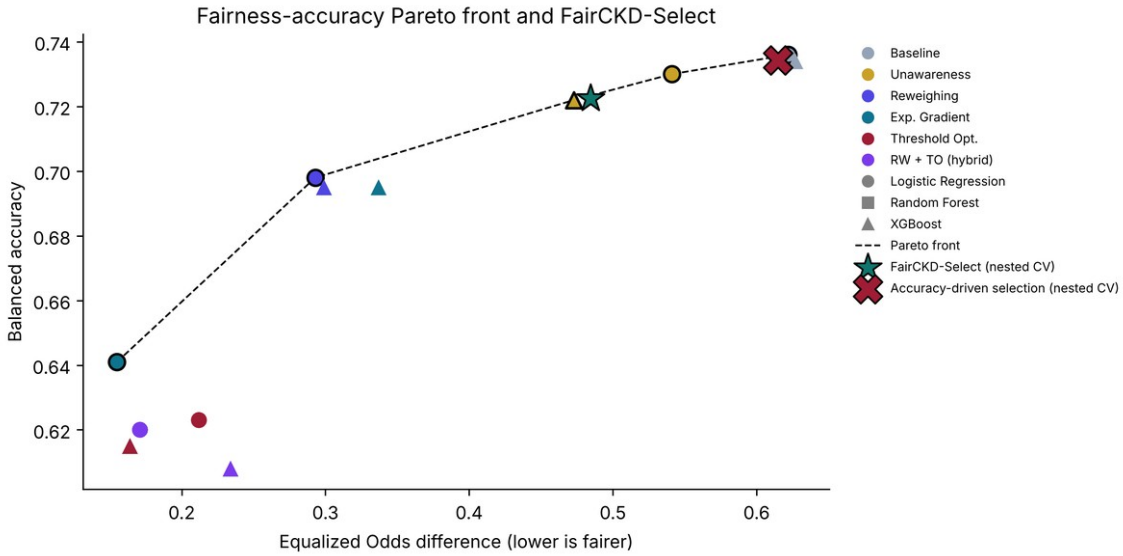
4b. FairCKD-Select (proposed method)

FairCKD-Select chooses, inside each training fold, the classifier and mitigation method with the smallest Equalized Odds gap among those whose balanced accuracy is within 3 points of the best unmitigated model (nested cross-validation). It is compared with conventional accuracy-driven selection.

Strategy	Accuracy	Balanced acc.	Recall	EOpp diff	EOdds diff	DP diff
FairCKD-Select	0.768 ± 0.005	0.723 ± 0.002	0.651 ± 0.002	0.484 ± 0.015	0.484 ± 0.015	0.443 ± 0.006
Accuracy-driven selection	0.761 ± 0.002	0.734 ± 0.004	0.692 ± 0.006	0.615 ± 0.010	0.615 ± 0.010	0.614 ± 0.011

Pipelines chosen by FairCKD-Select across outer folds

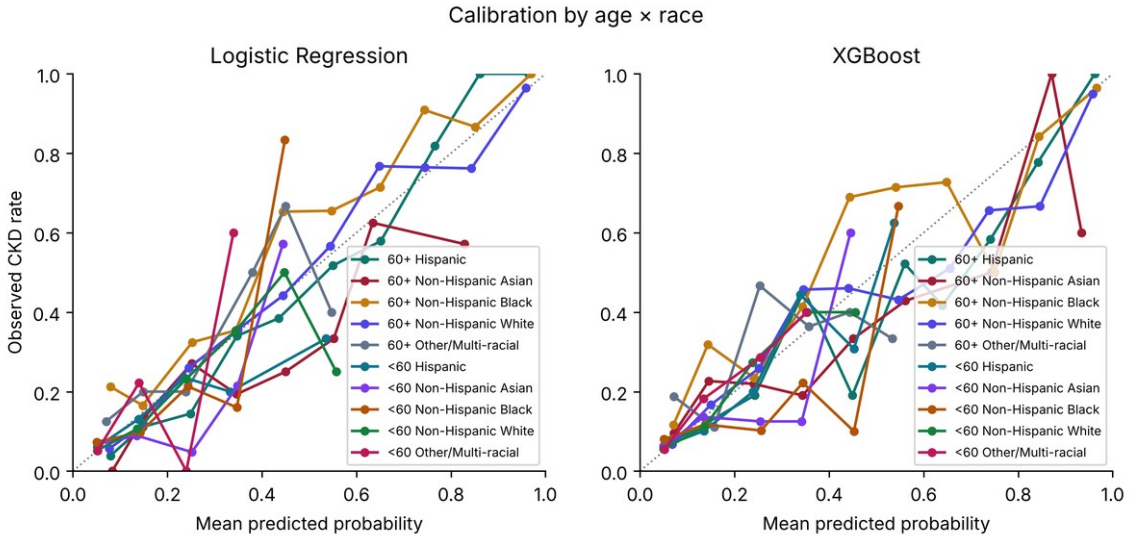
Model	Method	Times selected
Logistic Regression	Unawareness	1
XGBoost	Unawareness	9



4c. Calibration by group

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
60+ Hispanic	372	27.688	32.099	4.411	0.055	0.147	Logistic Regression
60+ Non-Hispanic Asian	199	24.623	30.937	6.314	0.081	0.168	Logistic Regression
60+ Non-Hispanic Black	423	41.371	34.068	-7.303	0.073	0.185	Logistic Regression
60+ Non-Hispanic White	763	36.042	35.806	-0.236	0.025	0.171	Logistic Regression
60+ Other/Multi-racial	70	34.286	31.084	-3.202	0.089	0.189	Logistic Regression
<60 Hispanic	832	9.495	9.303	-0.192	0.013	0.080	Logistic Regression
<60 Non-Hispanic Asian	510	9.608	10.63	1.023	0.038	0.076	Logistic Regression
<60 Non-Hispanic Black	698	10.888	10.981	0.092	0.038	0.089	Logistic Regression
<60 Non-Hispanic White	999	8.609	9.345	0.737	0.010	0.073	Logistic Regression
<60 Other/Multi-racial	197	11.675	9.915	-1.76	0.038	0.087	Logistic Regression
60+ Hispanic	372	27.688	30.668	2.98	0.062	0.151	XGBoost
60+ Non-Hispanic Asian	199	24.623	27.511	2.888	0.088	0.173	XGBoost
60+ Non-Hispanic Black	423	41.371	34.065	-7.306	0.097	0.184	XGBoost
60+ Non-Hispanic White	763	36.042	37.048	1.006	0.046	0.175	XGBoost
60+ Other/Multi-racial	70	34.286	30.741	-3.545	0.116	0.212	XGBoost

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
<60 Hispanic	832	9.495	9.475	-0.020	0.026	0.078	XGBoost
<60 Non-Hispanic Asian	510	9.608	9.285	-0.323	0.023	0.077	XGBoost
<60 Non-Hispanic Black	698	10.888	11.126	0.238	0.046	0.091	XGBoost
<60 Non-Hispanic White	999	8.609	8.991	0.382	0.016	0.074	XGBoost
<60 Other/Multi-racial	197	11.675	9.766	-1.909	0.019	0.082	XGBoost

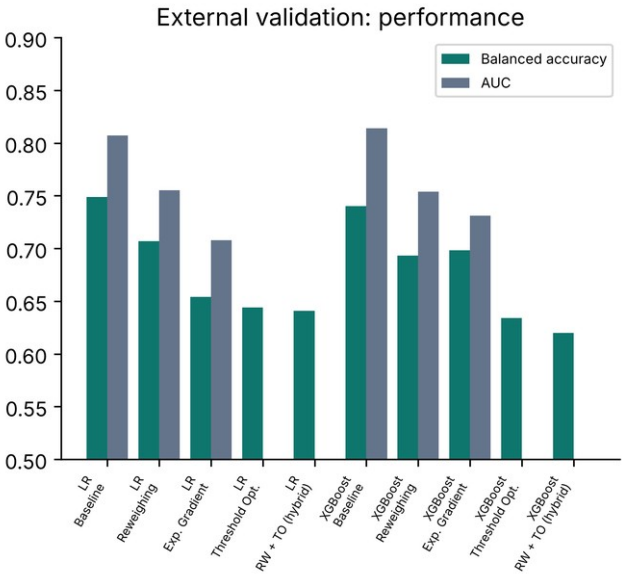


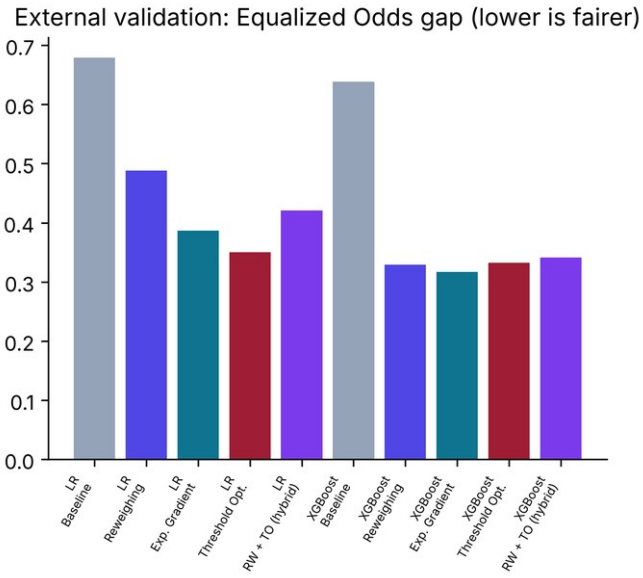
4d. Population-weighted estimates (NHANES survey weights)

ispanic]	W TPR [60+ Non-Hispanic Asian]	W TPR [60+ Non-Hispanic Black]	W TPR [60+ Non-Hispanic White]	W TPR [60+ Other/Multi-racial]	W TPR [<60 Hispanic]	W TPR [<60 Non-His
	0.834	0.871	0.919	0.891	0.276	0.306
	0.767	0.777	0.824	0.809	0.260	0.254
	0.807	0.858	0.907	0.951	0.356	0.364
	0.609	0.703	0.687	0.742	0.207	0.240
	0.666	0.665	0.684	0.745	0.325	0.259
	0.605	0.699	0.690	0.719	0.321	0.274
	0.743	0.815	0.846	0.863	0.291	0.293

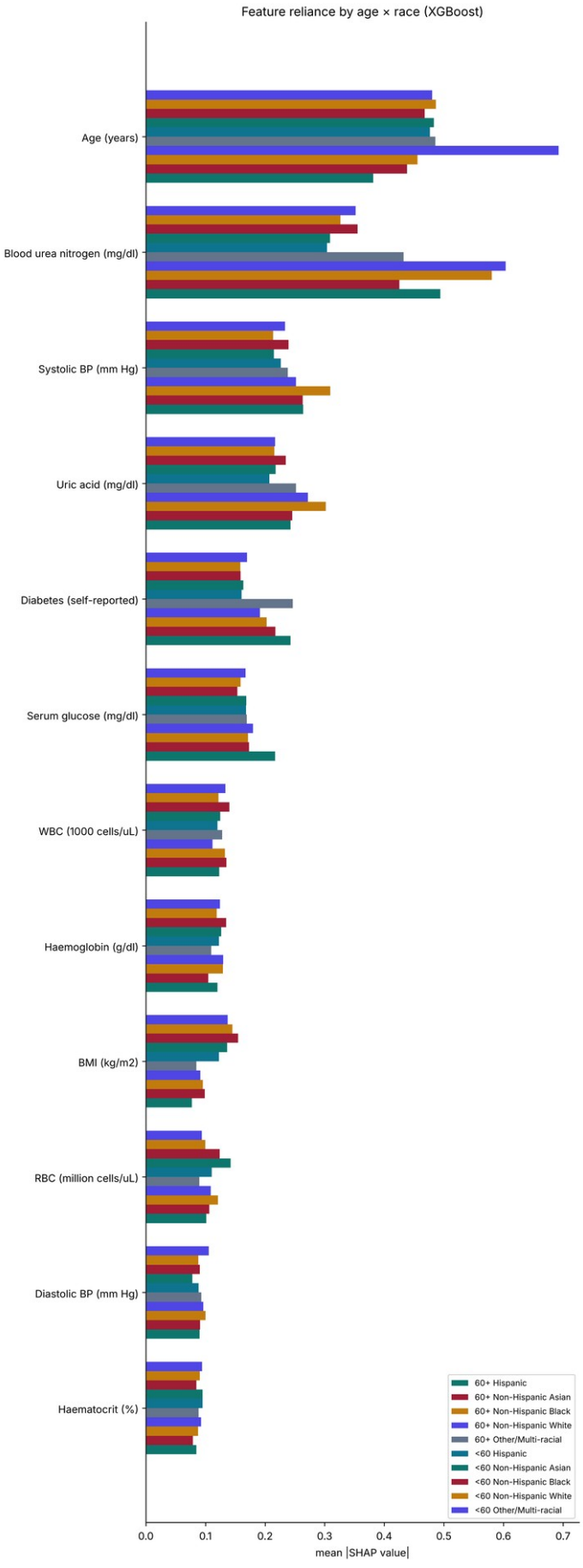
4e. External validation (trained on NHANES 2017–18, tested on NHANES 2015–16)

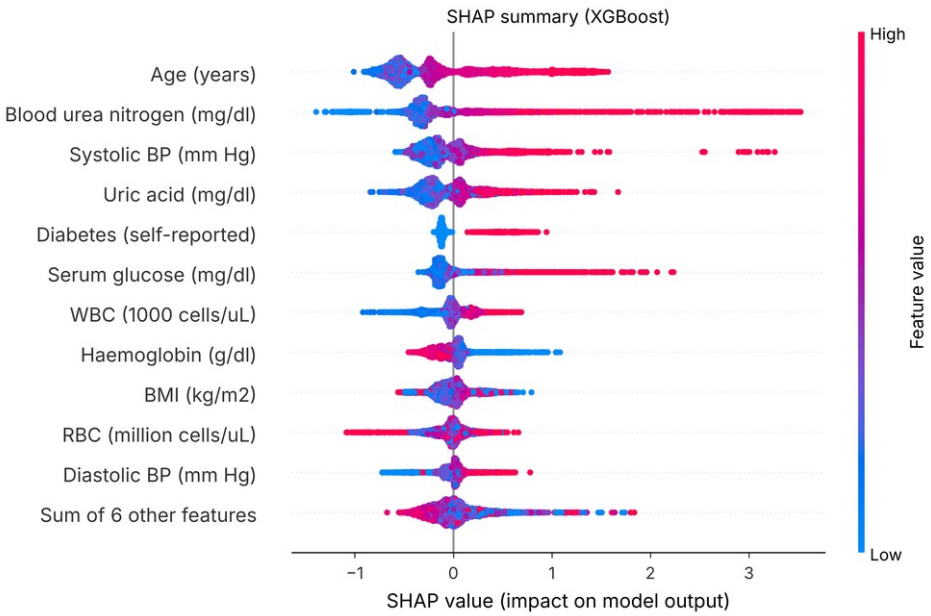
Model	Method	Accuracy	Balanced acc.	Recall	AUC	EOpp diff	EOdds diff
Logistic Regression	Baseline	0.784	0.749	0.696	0.807	0.679	0.679
Logistic Regression	Reweighing	0.743	0.707	0.655	0.755	0.488	0.488
Logistic Regression	Exp. Gradient	0.686	0.654	0.608	0.708	0.387	0.387
Logistic Regression	Threshold Opt.	0.809	0.644	0.400		0.350	0.350
Logistic Regression	RW + TO (hybrid)	0.676	0.641	0.589		0.421	0.421
XGBoost	Baseline	0.791	0.740	0.666	0.814	0.638	0.638
XGBoost	Reweighing	0.746	0.693	0.614	0.754	0.329	0.329
XGBoost	Exp. Gradient	0.800	0.698	0.547	0.731	0.317	0.317
XGBoost	Threshold Opt.	0.706	0.634	0.527		0.332	0.332
XGBoost	RW + TO (hybrid)	0.651	0.620	0.575		0.341	0.341





6. Explainability (SHAP)





Protocol: repeated stratified 5-fold cross-validation (stratified on label × group); preprocessing and mitigation fitted on training folds only; mean ± SD over repeats. This system is a research prototype and not a medical device.